

PrimeRx Flow - Pharma Alert to Prescription Analysis

This project is a Python + Pandas analysis on PrimeRx pharma data. The main idea is simple: connect clinical alerts with doctor prescription behavior and find where the business opportunity is getting missed.

Flow of the project:

```
Clinical Alerts -> Doctor Behavior -> Prescriptions -> Sales Strategy
```

Using alerts, sales and hospital affiliation data, I checked which doctors have high patient need, which doctors are not converting alerts into prescriptions, where competitors are winning, and which accounts need more field force focus.

Dataset

The dataset is stored in `data_proc.xlsx` and contains three sheets.

Alerts: clinical alert data for doctors

Columns: `Alert_ID`, `Alert_Date`, `HCP_ID`, `Lab_Result`

Sales: prescription transactions

Columns: `Prescription_ID`, `Prescription_Date`, `HCP_ID`, `Drug_Name`, `Drug_ID`, `Prescription_Volume`

Affiliation: doctor to hospital/account mapping

Columns: `HCP_ID`, `Account_ID`, `Account_Name`

Data Loading and Cleaning

First I loaded all sheets, cleaned the doctor IDs and converted date columns into datetime format so the analysis can be done monthly and at HCP level.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

FILE = "data_proc.xlsx"

alerts = pd.read_excel(FILE, sheet_name="Alerts")
sales = pd.read_excel(FILE, sheet_name="Sales")
affil = pd.read_excel(FILE, sheet_name="Affiliation")

alerts["HCP_ID"] = alerts["HCP_ID"].str.strip().str.upper()
sales["HCP_ID"] = sales["HCP_ID"].str.strip().str.upper()
affil["HCP_ID"] = affil["HCP_ID"].str.strip().str.upper()
```

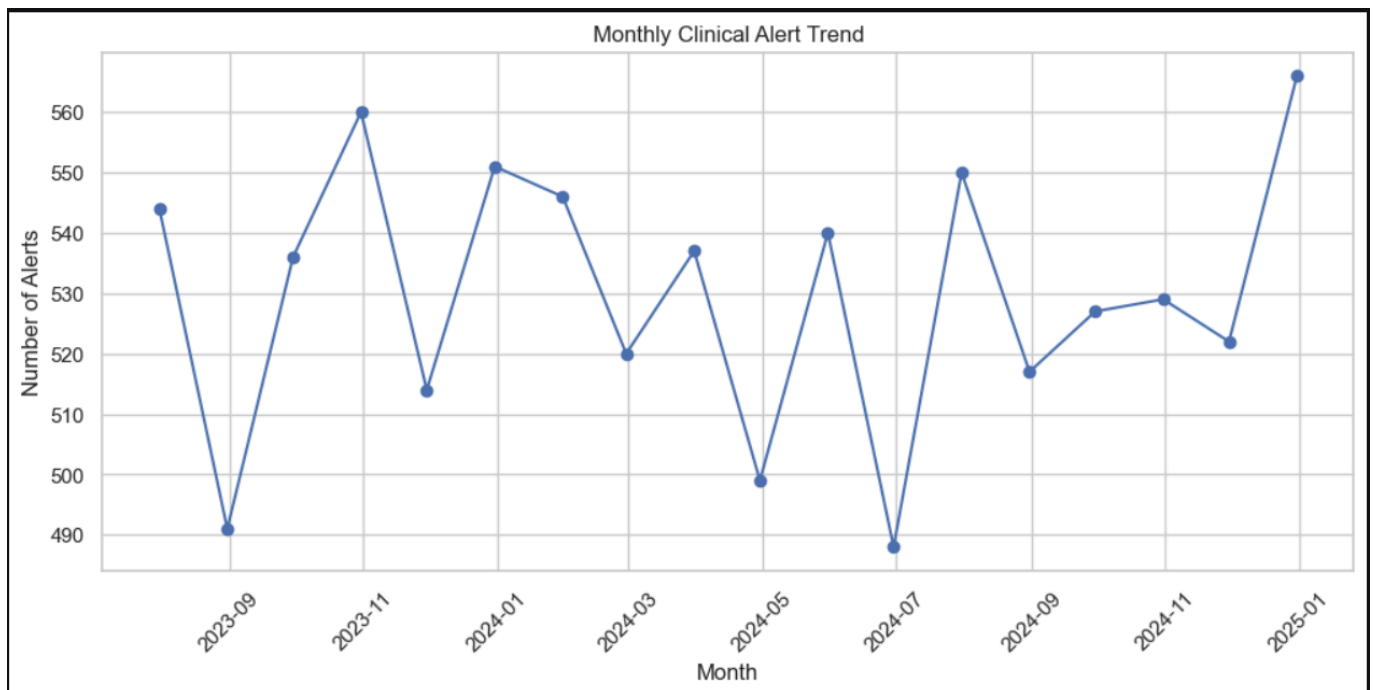
```
alerts["Alert_Date"] = pd.to_datetime(alerts["Alert_Date"])
sales["Prescription_Date"] = pd.to_datetime(sales["Prescription_Date"])
```

1. Monthly Alert Trend

Business question: are patient opportunities increasing over time?

```
alert_trend = alerts.set_index("Alert_Date").resample("M").size()

plt.figure(figsize=(12, 5))
plt.plot(alert_trend.index, alert_trend.values, marker="o")
plt.title("Monthly Clinical Alert Trend")
plt.xlabel("Month")
plt.ylabel("Alerts")
plt.savefig("one.png", bbox_inches="tight")
plt.show()
```



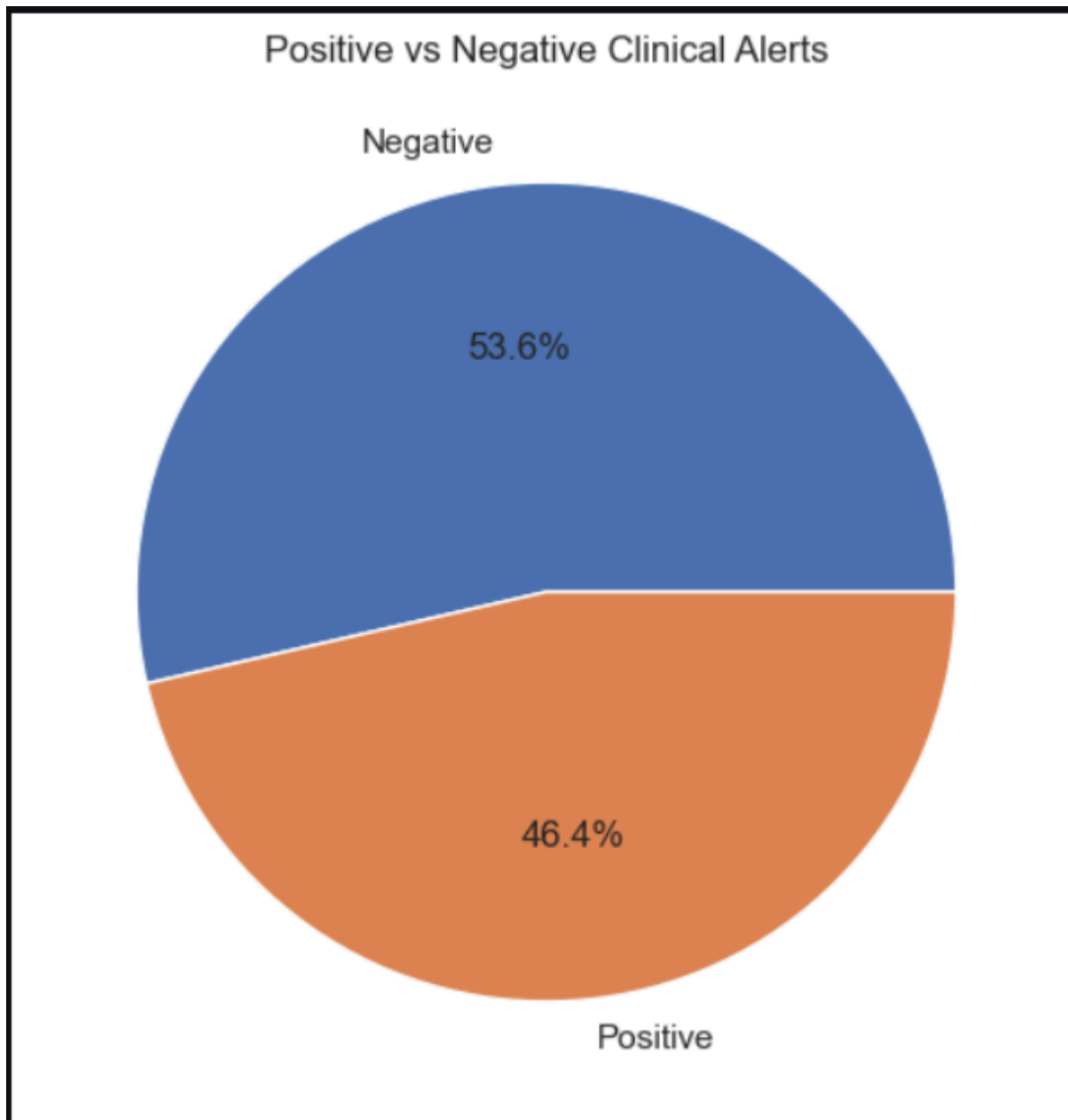
Insight: alerts are increasing, which means the treatment opportunity is also growing.

2. Positive vs Negative Alerts

Business question: how many alerts are actual treatment opportunities?

```
alert_type = alerts["Lab_Result"].value_counts()

plt.pie(alert_type, labels=alert_type.index, autopct="%1.1f%%")
plt.title("Positive vs Negative Alerts")
plt.savefig("two.png", bbox_inches="tight")
plt.show()
```



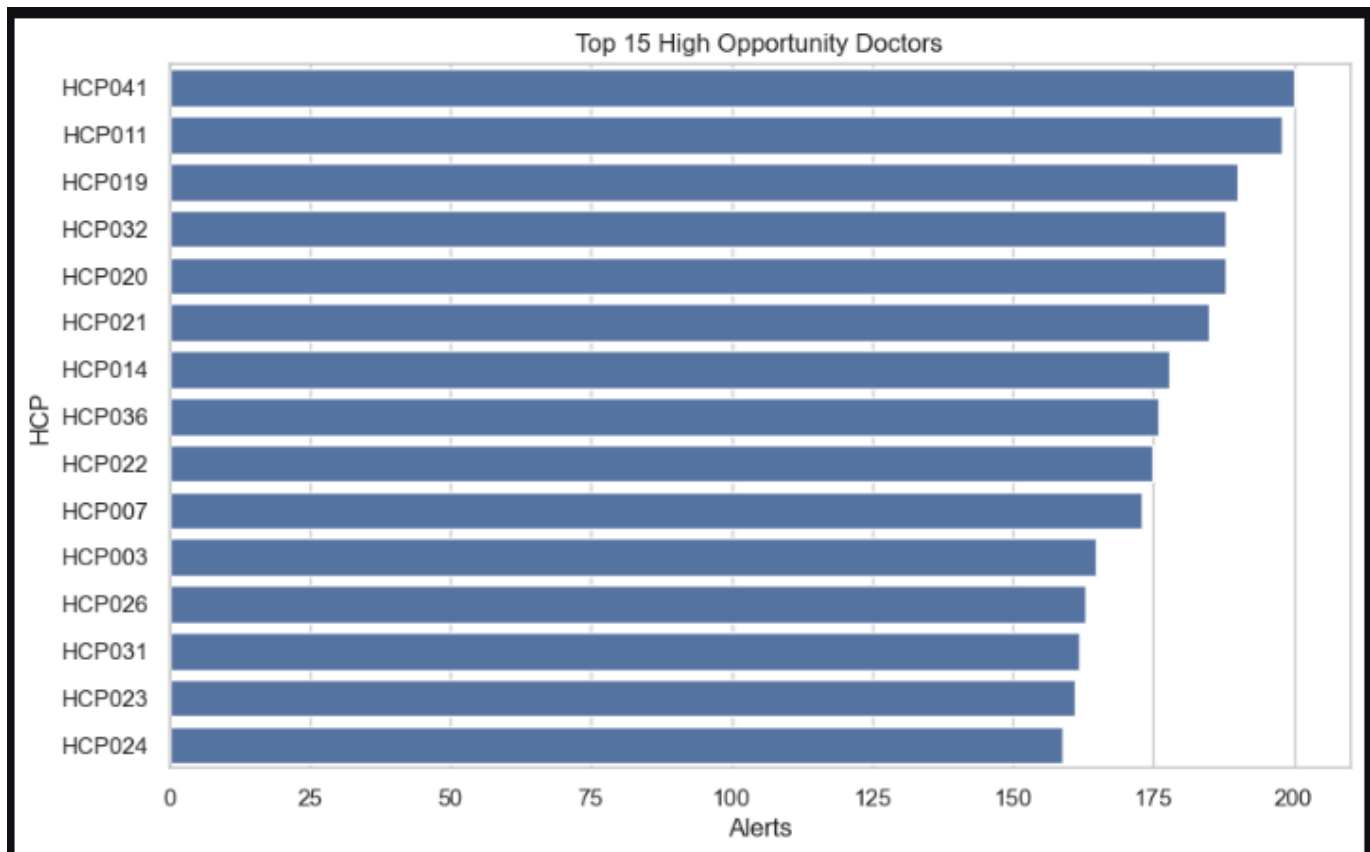
Insight: positive alerts show where patients may need treatment, so these alerts are more valuable for commercial action.

3. Top Doctors by Alert Volume

This finds the doctors with the highest number of alerts.

```
top_alerts = alerts.groupby("HCP_ID").size().sort_values(ascending=False).head(15)

sns.barplot(x=top_alerts.values, y=top_alerts.index)
plt.title("Top 15 High Opportunity HCPs")
plt.savefig("three.png", bbox_inches="tight")
plt.show()
```



Insight: these HCPs should be checked first because they are seeing more patient opportunities.

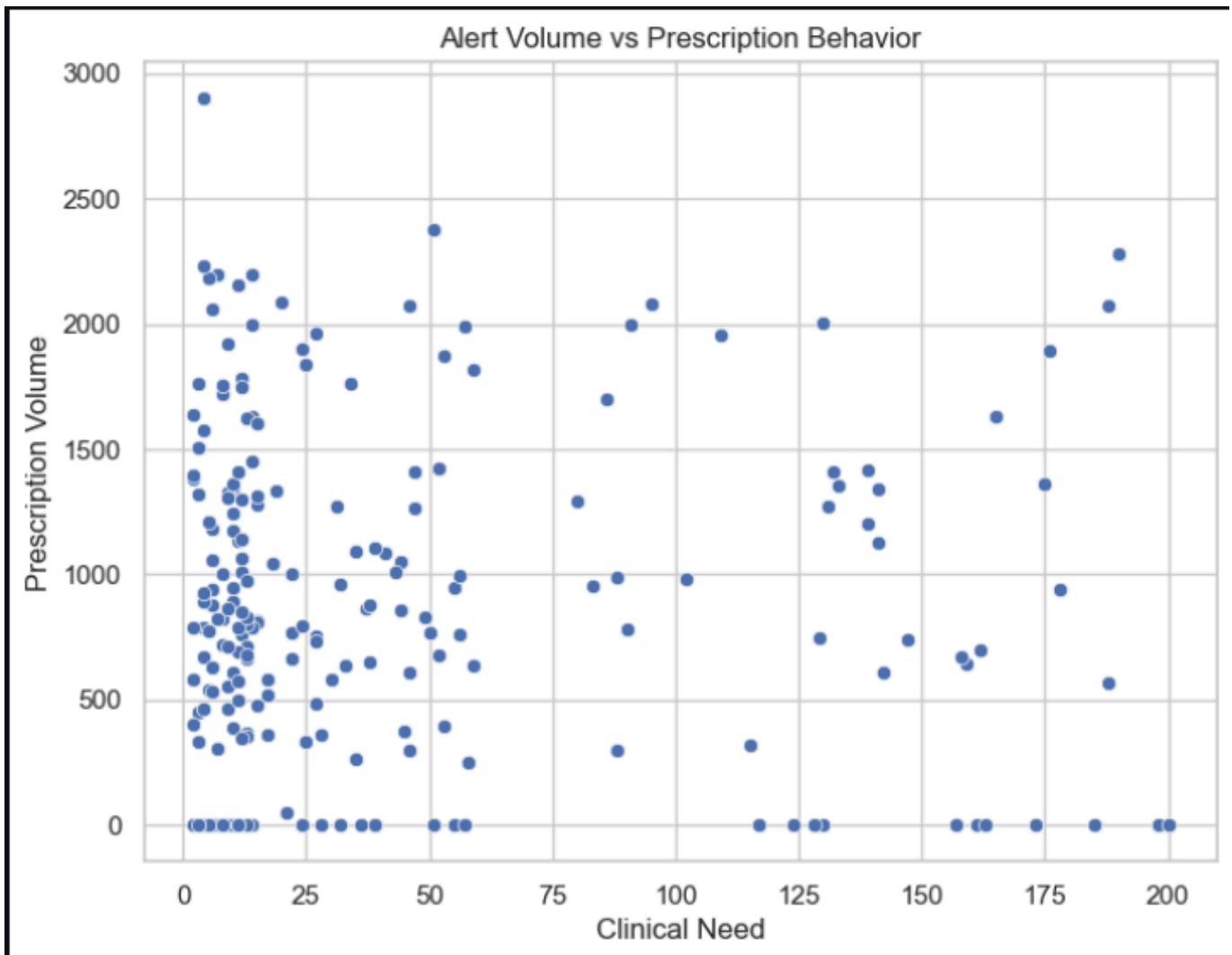
4. Alert vs Prescription Gap

This is the main opportunity view. It compares alert volume with actual prescription volume.

```
hcp_alerts = alerts.groupby("HCP_ID").size().reset_index(name="Alerts")
hcp_rx = sales.groupby("HCP_ID")
["Prescription_Volume"].sum().reset_index(name="Rx")

gap = hcp_alerts.merge(hcp_rx, on="HCP_ID", how="left")
gap["Rx"] = gap["Rx"].fillna(0)

sns.scatterplot(data=gap, x="Alerts", y="Rx")
plt.title("Alert Volume vs Prescription Behavior")
plt.savefig("four.png", bbox_inches="tight")
plt.show()
```



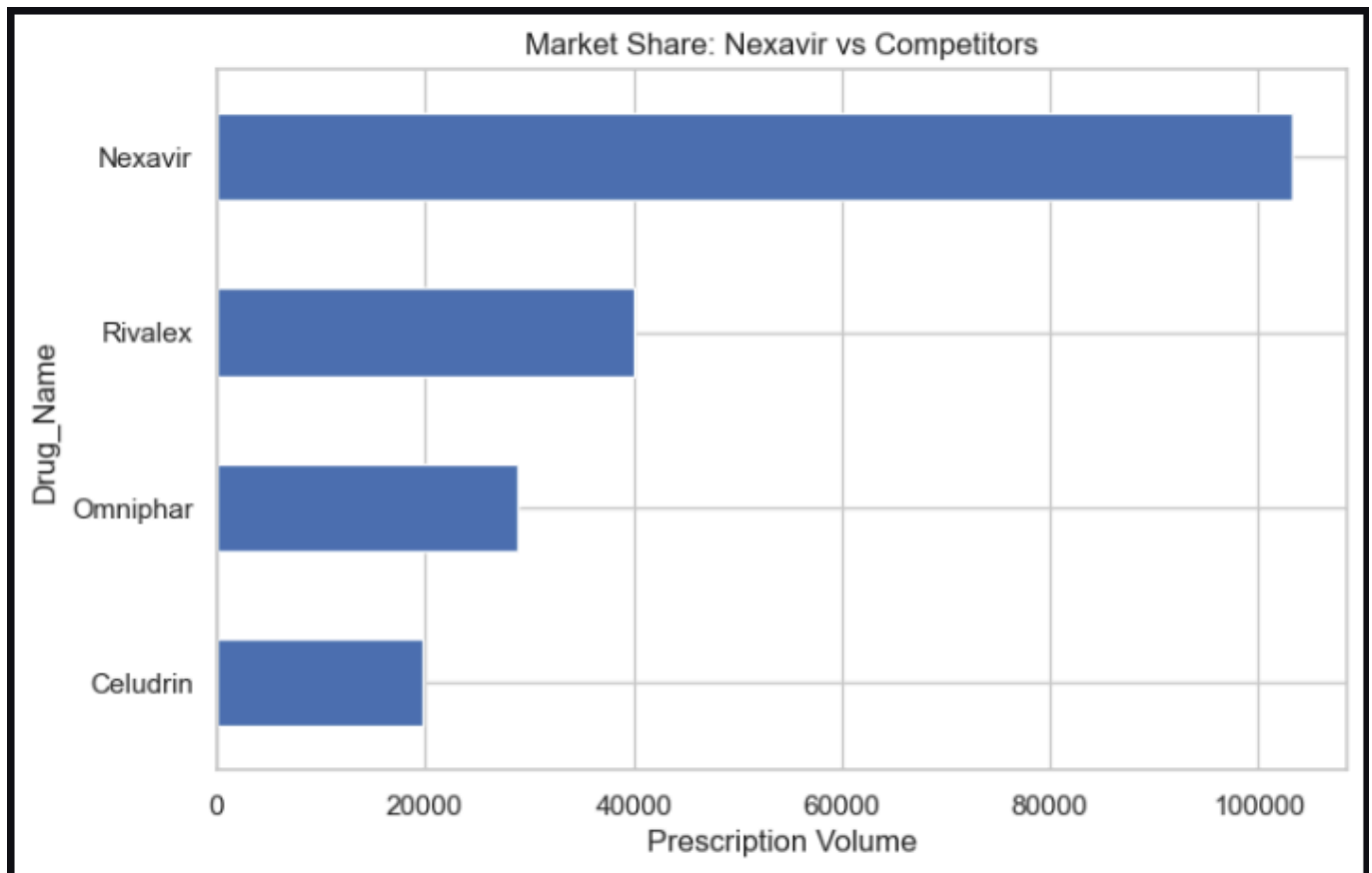
Insight: high alerts with low prescriptions means untapped opportunity. These doctors should be a priority for sales follow-up.

5. Competitor Leakage

This checks how Nexavir is performing against competitor drugs.

```
drug_share = sales.groupby("Drug_Name")["Prescription_Volume"].sum()

drug_share.plot(kind="barh")
plt.title("Market Share Analysis")
plt.savefig("five.png", bbox_inches="tight")
plt.show()
```



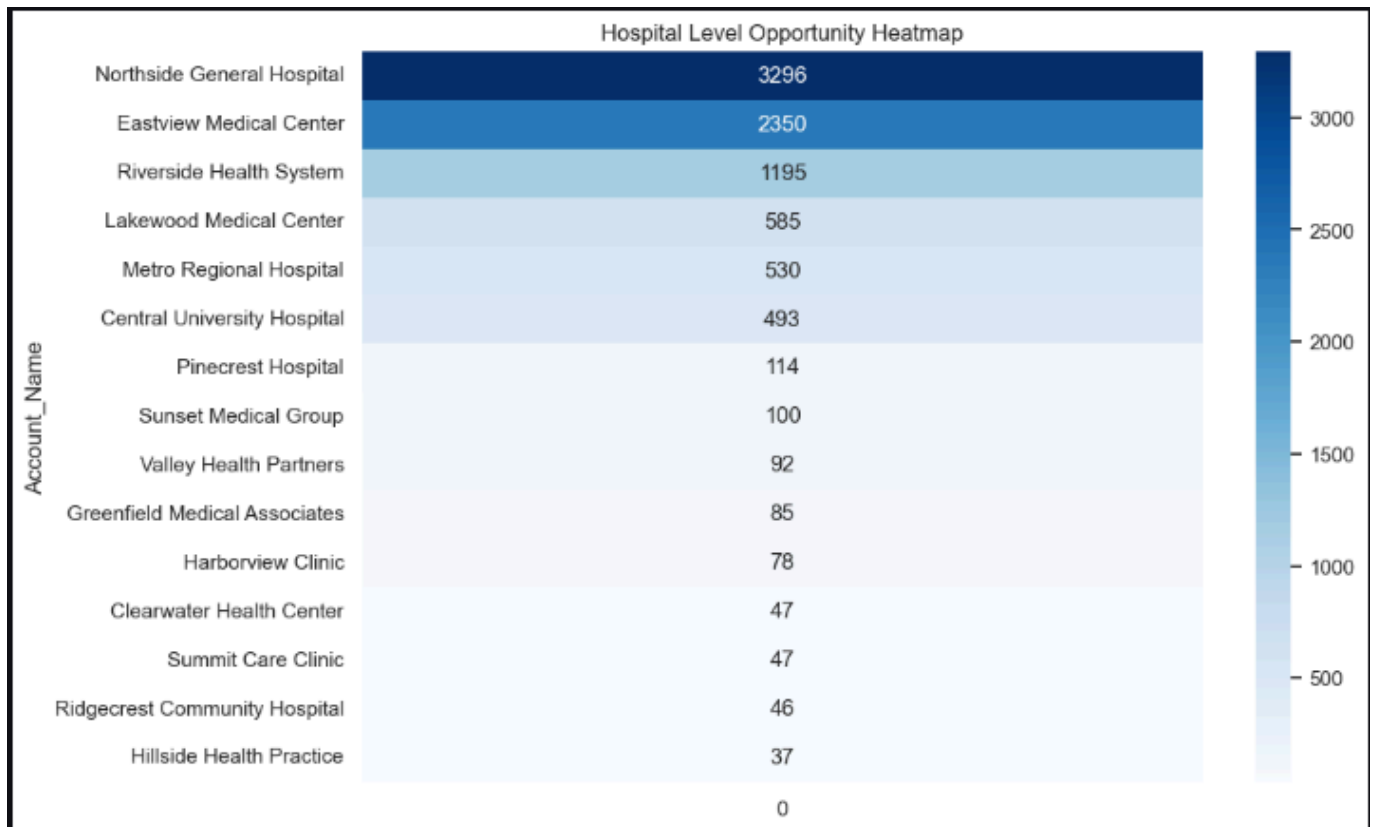
Insight: high competitor volume means doctors are treating patients, but not choosing Nexavir.

6. Account Opportunity Heatmap

This shows hospitals/accounts with the highest alert load.

```
account_alerts = alerts.merge(affil, on="HCP_ID")
heat =
account_alerts.groupby("Account_Name").size().sort_values(ascending=False).head(15
)

sns.heatmap(heat.to_frame(), annot=True)
plt.title("Hospital Opportunity Heatmap")
plt.savefig("six.png", bbox_inches="tight")
plt.show()
```



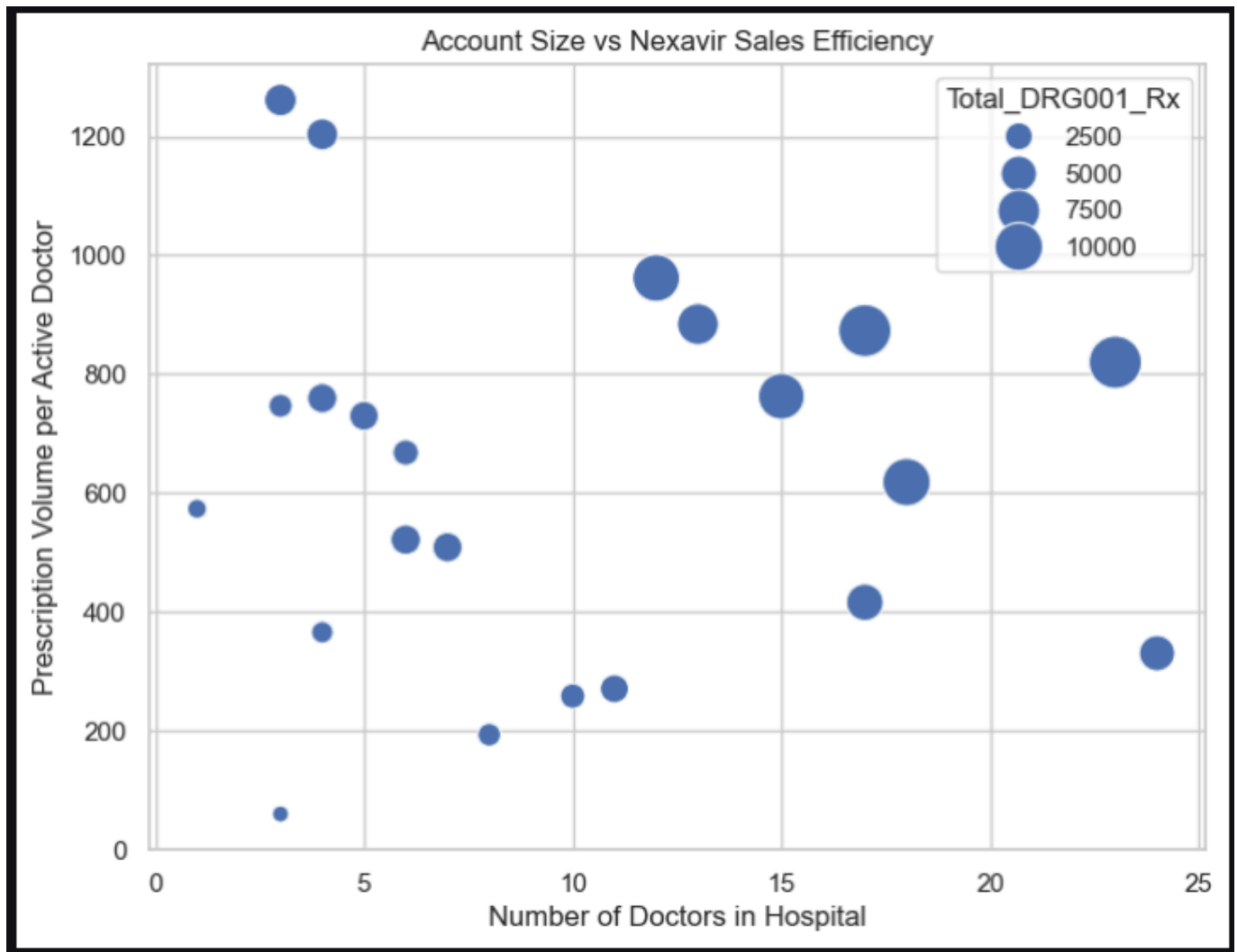
Insight: accounts with more alerts should get stronger account-level planning.

7. Account Size vs Sales Efficiency

This compares hospital size with Nexavir prescription efficiency.

```
sns.scatterplot(
    data=q5_active,
    x="Total_Doctors",
    y="Rx_Per_Active_Doctor",
    size="Total_DRG001_Rx"
)

plt.title("Hospital Size vs Nexavir Efficiency")
plt.savefig("seven.png", bbox_inches="tight")
plt.show()
```



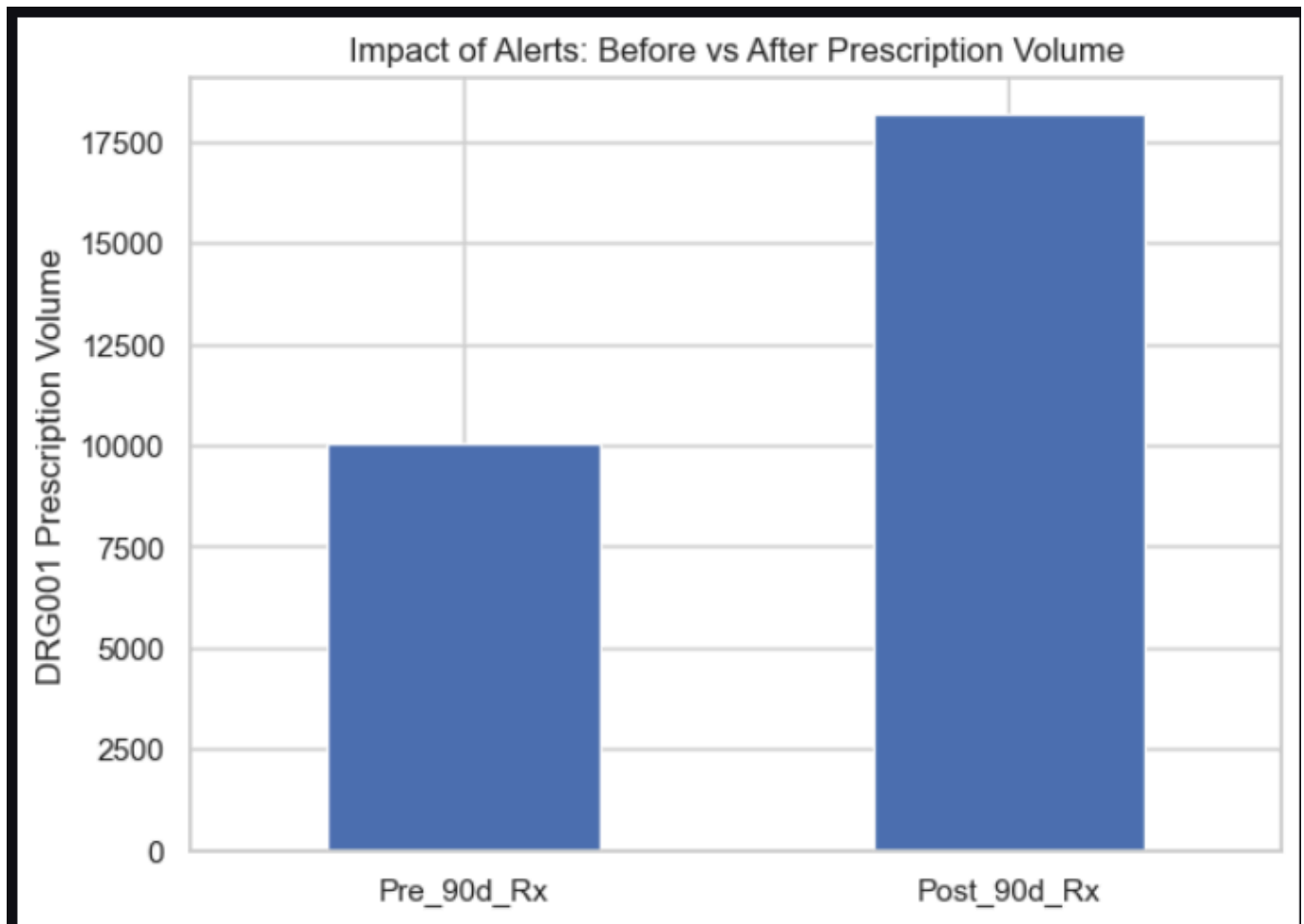
Insight: large hospitals with low Rx per doctor need better account strategy.

8. Before vs After Alert Lift

This checks whether alerts changed doctor behavior after they were generated.

```
lift_compare = q6[["Pre_90d_Rx", "Post_90d_Rx"]].sum()

lift_compare.plot(kind="bar")
plt.title("Before vs After Alert Impact")
plt.savefig("eight.png", bbox_inches="tight")
plt.show()
```

Insight: if post-alert prescriptions increase, alerts are helping drive action.

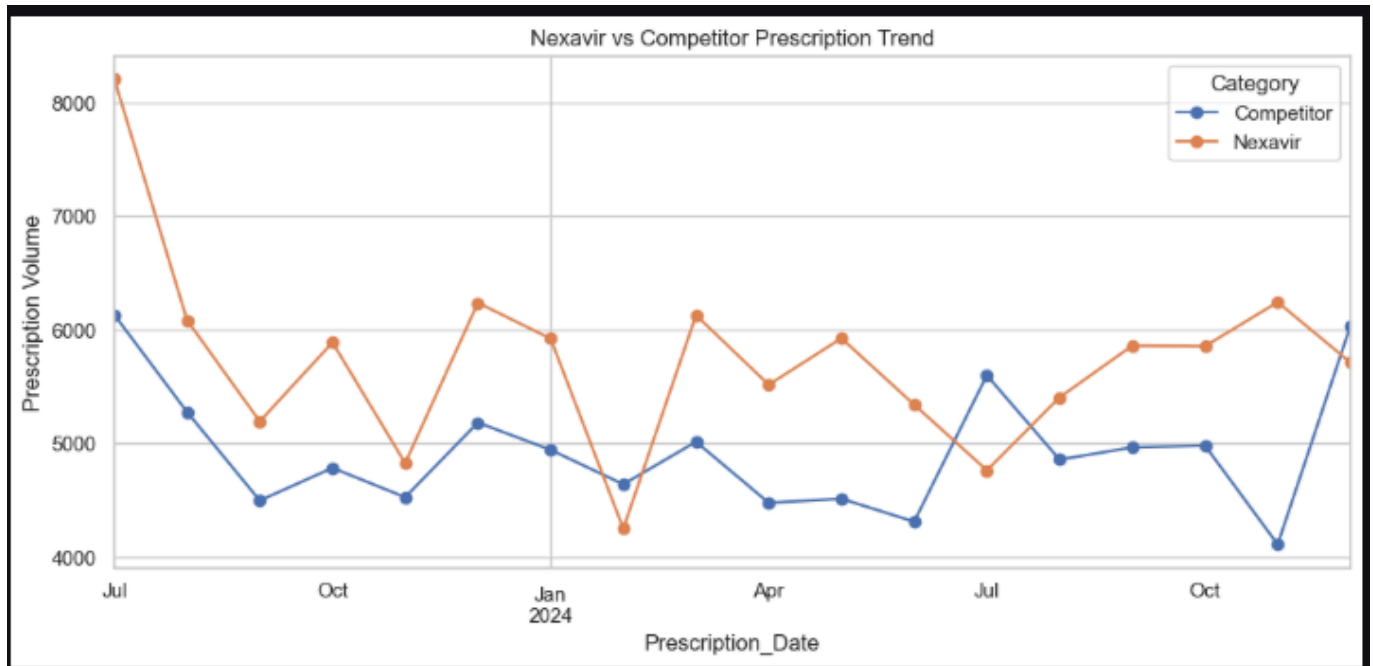
9. Nexavir vs Competitor Prescription Trend

Business question: is Nexavir gaining prescription volume over time compared to competitors?

```
temp = sales.copy()
temp["Category"] = temp["Drug_ID"].apply(lambda x: "Nexavir" if x == "DRG001" else
"Competitor")

trend = (
    temp.groupby([pd.Grouper(key="Prescription_Date", freq="M"), "Category"])
    ["Prescription_Volume"]
    .sum()
    .unstack()
)

trend.plot(figsize=(12, 5), marker="o")
plt.title("Nexavir vs Competitor Prescription Trend")
plt.xlabel("Month")
plt.ylabel("Prescription Volume")
plt.xticks(rotation=45)
plt.savefig("nine.png", bbox_inches="tight")
plt.show()
```



Insight: if Nexavir is increasing, adoption is improving. If competitors remain high, the team should target competitor-heavy HCPs and improve product positioning.

Final Business Recommendations

Target high-alert low-prescription doctors because they have patient need but low conversion.

Attack competitor leakage by focusing on doctors who prescribe competitor drugs instead of Nexavir.

Prioritize underperforming hospitals where many doctors or alerts exist but sales efficiency is low.

Invest more in new starters and growers because these segments are more likely to respond to field force action.

Segment	Priority
New Starters	High
Growers	High
Non Responders	Low

Tech Stack

- Python
- Pandas
- Matplotlib
- Seaborn
- Excel
- Jupyter Notebook

Conclusion

This project converts raw pharma alert and sales data into a clear commercial action plan. It shows where patients exist, which doctors are not converting, where competitors are winning, which accounts need focus, and how sales teams can prioritize outreach.